

Lecture 9: Processes on Networks I

9.1 Network Resilience and Percolation

Percolation: formation of a giant component when nodes/edges are randomly retained.

Site percolation: remove nodes (with edges).

Bond percolation: remove edges.

Percolation threshold for Erdős–Rényi (ER): average degree $z_c = 1$.

$z > 1 \Rightarrow$ giant component exists.

After random edge removal with probability p , new avg. degree: $z' = pz_0$.

Robustness:

Scale-free networks: robust to random failures, fragile to targeted attacks (removing hubs).

For scale-free with $\gamma \in (2, 3)$, $\langle k^2 \rangle \rightarrow \infty \Rightarrow$ no percolation threshold under random removal (always percolates).

9.2 Cascade Failures

Node i has load L_i and capacity $C_i = (1 + \alpha)L_i^{(0)}$, where α = tolerance parameter.

If load $> C_i$, node fails; load redistributed $\Rightarrow$ cascade.

Path efficiency: $e_{\text{path}} = \left(\sum_{\ell \in \text{path}} \frac{1}{e_\ell} \right)^{-1}$, $e_\ell \in [0, 1]$.

Network efficiency: average of most efficient generator $\rightarrow$ distribution paths.

9.3 Synchronization

Linearly coupled identical oscillators:

$$\frac{dx_i}{dt} = F(x_i) + \sigma \sum_{j \in V(i)} [H(x_i) - H(x_j)] \Rightarrow \text{Laplacian } L.$$

Master Stability Function (MSF):

$$\Lambda(\alpha) = \max_s (F'(s) + \alpha H'(s)).$$

Synchronized state stable iff $\Lambda(\sigma \lambda_i) \leq 0$ for all $i \geq 2$, where $0 = \lambda_1 < \lambda_2 \leq \dots \leq \lambda_N$ are Laplacian eigenvalues.

If $\Lambda(\alpha) < 0$ for $\alpha > \alpha_c$, then $\sigma > \alpha_c / \lambda_2$ ensures sync.

Typical sync interval: $\sigma \lambda_2 > \alpha_1$ and $\sigma \lambda_{\max} < \alpha_2$.

Kuramoto model:

$$\frac{dx_i}{dt} = \omega_i + K \sum_{j \sim i} \sin(x_i - x_j).$$

Order parameter: $r = \left| \frac{1}{N} \sum_j e^{ix_j} \right|$, $r \approx 0$ (incoherent), $r \rightarrow 1$ (sync).

Critical coupling for heterogeneous network:

$$K_c \propto \frac{\langle k \rangle}{\langle k^2 \rangle}.$$

Alternatively, $K_c \propto 1/\lambda_{\max}(A)$, where A = adjacency matrix.

Higher degree heterogeneity $\Rightarrow$ easier synchronization.

9.4 Network Search

Search strategies:

Broadcast: shortest path, high traffic.

Random walk: low cost, long path.

Degree-biased: route to high-degree nodes.

Greedy with local info (e.g., geography, hierarchy): achieves near-shortest paths (e.g., “six degrees”).

In social networks, hierarchical grouping enables faster decentralized search than pure degree-based routing.

Lecture 10: Processes on Networks II

10.1 Epidemic Models Overview

Full mixing: homogeneous contact; **Network:** contact via edges (more realistic).

Key question: Will an outbreak become an epidemic?

10.2 SI Model

States: S (susceptible), I (infected, never recovers).

Mean-field:

$$\frac{di}{dt} = \beta(1-i)i, \quad i(t) = \frac{i_0 e^{\beta t}}{1 - i_0 + i_0 e^{\beta t}}.$$

On network (early stage):

$$\frac{di}{dt} = \beta A i \Rightarrow i(t) \sim e^{\beta \kappa_1 t},$$

where κ_1 = largest eigenvalue of A .

Time scale (degree-based approx):

$$\tau = \frac{\langle k \rangle}{\beta(\langle k^2 \rangle - \langle k \rangle)}.$$

Large $\langle k^2 \rangle$ (e.g., scale-free) $\Rightarrow$ fast outbreak.

10.3 SIR Model

States: $S \rightarrow I \rightarrow R$ (recovered/removed, immune).

Mean-field:

$$\frac{di}{dt} = \beta i s - \mu i, \quad \frac{ds}{dt} = -\beta i s, \quad \frac{dr}{dt} = \mu i.$$

Basic reproduction number:

$$R_0 = \frac{\beta}{\mu}.$$

Epidemic threshold: $R_0 = 1$.

Final size r_∞ satisfies:

$$r_\infty = 1 - e^{-R_0 r_\infty}.$$

Percolation equivalence: each edge is “open” with probability

$$p = \frac{\beta}{\beta + \mu}.$$

Infected cluster = percolation cluster from initial seeds.

Network threshold (no degree correlation):

$$\frac{\beta}{\mu} \geq \frac{\langle k \rangle}{\langle k^2 \rangle - \langle k \rangle} \Leftrightarrow R_0^{\text{eff}} = \frac{\beta}{\mu} \cdot \frac{\langle k^2 \rangle - \langle k \rangle}{\langle k \rangle} > 1.$$

For scale-free ($2 < \gamma \leq 3$), $\langle k^2 \rangle \rightarrow \infty \Rightarrow$ threshold $\rightarrow 0$.

10.4 SIS Model

States: $S \leftrightarrow I$ (no immunity).

Mean-field:

$$\frac{di}{dt} = \beta i(1-i) - \mu i, \quad R_0 = \frac{\beta}{\mu}.$$

Steady state:

$$i^* = \begin{cases} 0, & R_0 \leq 1, \\ 1 - \frac{1}{R_0}, & R_0 > 1. \end{cases}$$

On network: early dynamics governed by $\lambda = \beta/\mu$, threshold

$$\lambda_c = \frac{1}{\kappa_1}, \quad \text{or} \quad \frac{\beta}{\mu} \geq \frac{\langle k \rangle}{\langle k^2 \rangle - \langle k \rangle}.$$

10.5 SIRS Model

States: $S \rightarrow I \rightarrow R \rightarrow S$ (temporary immunity, rate δ).

Equations:

$$\frac{di}{dt} = \beta i s - \mu i, \quad \frac{ds}{dt} = \delta r - \beta i s, \quad \frac{dr}{dt} = \mu i - \delta r.$$

No analytic solution.

For $\delta > 0$, can exhibit sustained oscillations (limit cycles) when $R_0 > 1$.

10.6 Key Comparisons

R_0 in homogeneous network: $R_0 = (\beta/\mu)\langle k \rangle$.
 R_0 in heterogeneous network: effective R_0 scaled by $\langle k^2 \rangle / \langle k \rangle$.
Scale-free networks: extremely vulnerable to epidemics (threshold ≈ 0).

Essential Formulas Summary

ER percolation threshold: $z_c = 1$

Epidemic threshold (heterogeneous): $\frac{\beta}{\mu} \geq \frac{\langle k \rangle}{\langle k^2 \rangle - \langle k \rangle}$

SI growth rate: $\sim e^{\beta \kappa_1 t}$

SIS steady infection: $i^* = 1 - 1/R_0 \ (R_0 > 1)$

Kuramoto sync threshold: $K_c \propto \langle k \rangle / \langle k^2 \rangle$

Cascade capacity: $C_i = (1 + \alpha)L_i^{(0)}$